

AWS EC2 Deployment Guide

Sameer AI Platform — Nuh Bot

Project	sameer-ai-platform
Owner	Sameer Shaik
Company	N.S (Nadeem Shaik)
AI Model	Nuh 1.0 (qwen2.5:1.5b)
Region	Asia Pacific — Sydney (ap-southeast-2)
Instance	m7i-flex.large (2 vCPU, 8GB RAM)
OS	Ubuntu 26.04 LTS
Date	June 2026

This document covers the complete step-by-step process of deploying the Nuh AI WhatsApp bot platform on AWS EC2, from account setup to running containers.

1. System Architecture

Your platform runs 4 Docker containers on a single EC2 instance:

Container	Port	Purpose
ai-gateway	8080	Spring Boot — connects WhatsApp bot to Nuh AI
whatsapp-bot	—	Node.js Baileys — handles WhatsApp messages
python-service	8000	FastAPI — manages mute/archive/block logic
open-webui	3000	Web UI dashboard for Nuh model

Flow: WhatsApp → Node.js bot → Python (block check) → Spring :8080 → Ollama (Nuh) → Reply

2. EC2 Instance Configuration

2.1 Instance Details

Setting	Value	Reason
Name	Learn_from_Nuh	Your project name
AMI	Ubuntu 26.04 LTS	Free tier eligible, stable
Instance Type	m7i-flex.large	2 vCPU, 8GB RAM — enough for all containers
Storage	30 GiB gp3	Fast SSD, ~\$2.40/month
Key Pair	sameer-nuh-key.pem	RSA key for SSH access

2.2 Security Group Rules

Rule	Port	Protocol	Source	Purpose
SSH	22	TCP	My IP only	Admin SSH access
Spring Gateway	8080	TCP	My IP only	AI API endpoint
Python Service	8000	TCP	My IP only	Block/mute service
Open WebUI	3000	TCP	My IP only	Web dashboard

Warning: Never set SSH source to 0.0.0.0/0 (Anywhere). Always use My IP for security.

2.3 Storage Configuration

Setting	Value
Size	30 GiB
Type	gp3 (General Purpose SSD)
IOPS	3000 (default)
Throughput	125 MB/s (default)
Encryption	Not encrypted (fine for dev)
File System	None

2.4 Advanced Details

Setting	Value
Termination Protection	Enable — prevents accidental deletion
Stop Protection	Enable — prevents accidental stop
Shutdown Behavior	Stop (not Terminate)
Purchasing Option	On-Demand — pay per hour, stop anytime
File Systems	None — single server, no shared storage needed
User Data	Optional startup script (see Section 4)

3. Key Pair & SSH Access

3.1 Key Pair Created

Key pair name: **sameer-nuh-key**

Type: RSA | Format: .pem | Downloaded to: ~/Downloads/sameer-nuh-key.pem

Warning: Save sameer-nuh-key.pem safely — you can NEVER download it again from AWS!

3.2 Move Key to Safe Location

```
cp ~/Downloads/sameer-nuh-key.pem ~/Documents/sameer-nuh-key.pem
```

```
chmod 400 ~/Documents/sameer-nuh-key.pem
```

3.3 SSH Into Your EC2 Instance

After launching, find your Public IP in EC2 Console → Instances → your instance.

```
ssh -i ~/Documents/sameer-nuh-key.pem ubuntu@YOUR_EC2_PUBLIC_IP
```

Note: Replace YOUR_EC2_PUBLIC_IP with the actual IP shown in EC2 console.

4. Server Setup After Launch

4.1 Update System

```
sudo apt update && sudo apt upgrade -y
```

4.2 Install Docker

```
sudo apt install docker.io docker-compose-plugin -y
```

```
sudo usermod -aG docker ubuntu
```

```
newgrp docker
```

```
docker --version
```

4.3 Install Ollama

```
curl -fsSL https://ollama.com/install.sh | sh
```

```
sudo systemctl start ollama
```

```
sudo systemctl enable ollama
```

4.4 Configure Ollama for Docker

```
sudo mkdir -p /etc/systemd/system/ollama.service.d
```

```
sudo tee /etc/systemd/system/ollama.service.d/override.conf << 'EOF' [Service]
Environment="OLLAMA_HOST=0.0.0.0" EOF
```

```
sudo systemctl daemon-reload && sudo systemctl restart ollama
```

4.5 Create Nuh Model

```
cat > /tmp/Modelfile << 'EOF' FROM qwen2.5:1.5b SYSTEM "" You are Nuh, an AI
assistant. Version: Nuh 1.0 Created by: Sameer Shaik Company: N.S Keep replies short.
Plain text only. "" PARAMETER temperature 0.7 EOF
```

```
ollama create Nuh -f /tmp/Modelfile
```

```
ollama list
```

5. Deploy Your Project

5.1 Copy Project to EC2

From your local machine (not EC2), run:

```
scp -i ~/Documents/sameer-nuh-key.pem -r ~/Desktop/sameer-ai-platform  
ubuntu@YOUR_EC2_IP:~/
```

5.2 Build Spring Jar on EC2

```
sudo apt install openjdk-21-jdk maven -y
```

```
cd ~/sameer-ai-platform/services/ai-gateway
```

```
mvn clean package -DskipTests -q
```

5.3 Start Everything

```
cd ~/sameer-ai-platform
```

```
docker compose up -d
```

```
docker compose ps
```

5.4 Scan WhatsApp QR Code

```
docker compose logs -f whatsapp-bot
```

A QR code will appear in the logs. Scan it with WhatsApp on your phone:

WhatsApp → Settings → Linked Devices → Link a Device → Scan QR

6. Cost & Instance Management

6.1 Cost Breakdown

Resource	Cost	Notes
m7i-flex.large running	~\$0.12/hr	~\$86/month if 24/7
30 GiB gp3 storage	~\$2.40/month	Charged even when stopped
Public IP (Elastic)	\$0.005/hr	Only when instance stopped
Data transfer	First 100GB free	Outbound to internet

6.2 Stop/Start to Save Money

State	Compute Cost	Storage Cost	Use when
Running	\$0.12/hr	\$0.08/GB/mo	Actively using/testing
Stopped	FREE	\$0.08/GB/mo	Not using — saves most cost
Terminated	FREE	FREE	Deleting forever — careful!

Warning: NEVER click Terminate unless you want to delete everything permanently. Stop is safe.

6.3 Daily Commands

```
# Start everything cd ~/sameer-ai-platform && docker compose up -d
```

```
# Stop everything docker compose down
```

```
# Check status docker compose ps
```

```
# View bot logs docker compose logs -f whatsapp-bot
```

```
# View all logs docker compose logs -f
```

7. Troubleshooting

Problem	Cause	Fix
Cannot SSH	Wrong IP or key	Check EC2 public IP, use chmod 400 on .pem
Bot not connecting	Spring not started	Check docker compose ps, logs ai-gateway
Ollama not responding	Not listening on 0.0.0.0	Check override.conf, restart ollama
Port not accessible	Security group	Add port rule in EC2 Security Groups
Out of memory	Too many containers	Stop open-webui if not needed
QR code not showing	Auth folder issue	Delete auth folder, restart whatsapp-bot

8. Security Best Practices

1. **Keep .pem file safe** — never share it, never upload to GitHub
2. **My IP only** — all security rules limited to your IP address
3. **Termination protection ON** — prevents accidental deletion
4. **Stop when not using** — reduces attack surface and saves cost
5. **Tavily API key** — keep it in environment variables, not in code
6. **Update regularly** — run `sudo apt update` monthly on your server